

Exponential Equations – Change of Base

Algebra 1 (US Common Core) | Solve without logarithms

Method

Rewrite both sides of the equation as powers of the *same base*, then set the exponents equal to each other and solve. For example, since $9 = 3^2$ and $27 = 3^3$, an equation like $3^{2n-1} = 27$ becomes $3^{2n-1} = 3^3$, so $2n - 1 = 3$.

1 Practice Problems

1. $3^{2n-1} = 27$

2. $\left(\frac{1}{9}\right)^{-2x+1} = 27^{2x-1}$

3. $8^{3x+2} = \left(\frac{1}{2}\right)^{-4x}$

4. $5^{-2k} = 5^{-2k}$

5. $\left(\frac{1}{9}\right)^{-p-2} = 81^{p-1}$

6. $\left(\frac{1}{2}\right)^{-4n} = 32$

7. $9^{-x+2} = 9^{2x-4}$

8. $\left(\frac{1}{64}\right)^{-v-2} = 16^{3v}$

9. $3^{4x} = 3^{3-x}$

10. $4^{k+2} = 64$

11. $27^{-3b} = 1$

12. $2^{-4p} = 16$

13. $16^{b-2} = \left(\frac{1}{4}\right)^{-2b}$

14. $64^{-2x+1} \cdot 16^{2x+1} = 16^{-x}$

15. $9^{-2n-2} = \frac{1}{81}$

16. $27^{2x+1} = 9^{3x}$

17. $16^x \cdot \left(\frac{1}{4}\right)^{-x-2} = \left(\frac{1}{4}\right)^{-x}$

18. $3^{4k} = 3^8$

19. $\frac{243^{2x}}{9} = 27$

20. $5^{-4x} = 5^{-x}$

21. $\frac{9^{-x}}{27^{-2x}} = 9^{-2x-4}$

22. $27^{2a+2} \left(\frac{1}{9}\right) = 81$

23. $9 \cdot 27^{2-3r} = \left(\frac{1}{3}\right)^{2r}$

24. $24 \cdot \left(\frac{1}{16}\right)^x = 16^{x-2}$

25. $64^{-2n-1} \cdot 16^{-3n} = \frac{1}{64}$

26. $4^b \cdot 16^{-2b} = 1$

27. $\frac{1}{81} \cdot 27^{2-2n} = 9$

28. $125 \cdot 25^{3n} = \left(\frac{1}{125}\right)^{-n}$

29. $4^{2-2k} \cdot 16^{3k} = 8^{2k}$

30. $\frac{1}{64} \cdot \left(\frac{1}{16}\right)^{-n} = 16^{-2n}$

31. $3^{2x+1} = 128^{x-1}$

32. $\left(\frac{1}{8}\right)^{2x-3} = 16^{x+2}$

33. $3^{2x+1} \cdot 9^{x-1} = 27^{x+2}$

34. $\frac{8^{2x+1}}{4^{x-1}} = 32$

35. $\left(\frac{1}{27}\right)^{-x+1} = 81^{2x-1}$

36. $100^{2x-1} = 1000^{x+1}$

37. $25^{x-2} \cdot 5^{3x} = 125^{x+1}$

38. $\frac{216^x}{36^{2x-1}} = 6^3$

39. $7^{3x-1} = 343$

40. $\left(\frac{1}{5}\right)^{2x-4} = 625^{x+1}$

2 Answer Key

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|-----------------------|-----------------------|------------------------|
| 1. $\{2\}$ | 15. $\{0\}$ | 29. $\{-2\}$ |
| 2. $\{\frac{1}{2}\}$ | 16. No solution | 30. $\{\frac{1}{2}\}$ |
| 3. $\{-2\}$ | 17. $\{-1\}$ | 31. $\{6\}$ |
| 4. All real numbers | 18. $\{2\}$ | 32. $\{\frac{1}{10}\}$ |
| 5. $\{2\}$ | 19. $\{\frac{1}{2}\}$ | 33. $\{7\}$ |
| 6. $\{\frac{5}{4}\}$ | 20. $\{0\}$ | 34. $\{0\}$ |
| 7. $\{2\}$ | 21. $\{-1\}$ | 35. $\{\frac{1}{5}\}$ |
| 8. No solution | 22. $\{0\}$ | 36. $\{5\}$ |
| 9. $\{\frac{3}{5}\}$ | 23. $\{\frac{8}{7}\}$ | 37. $\{\frac{7}{2}\}$ |
| 10. $\{1\}$ | 24. $\{\frac{3}{2}\}$ | 38. $\{-1\}$ |
| 11. $\{0\}$ | 25. $\{0\}$ | 39. $\{\frac{4}{3}\}$ |
| 12. $\{-1\}$ | 26. $\{0\}$ | 40. $\{0\}$ |
| 13. No solution | 27. $\{0\}$ | |
| 14. $\{\frac{5}{3}\}$ | 28. $\{-1\}$ | |